

Project 3 – Cloud Connected Security Node using ESP32

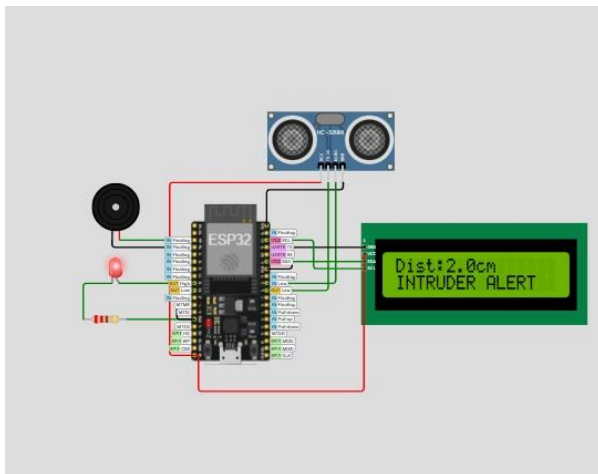
Objective:

The objective of this project is to design a cloud – connected security system using ESP32 that detects intruders and sends alerts through cloud communication. The system uses an ultrasonic sensor to measure distance and identify suspicious movement.

Components used:

- ESP32
- HC – SR04 Ultrasonic Sensor
- LED
- 220Ω Resistor
- Piezo Buzzer
- 16 x 2 I2C LCD Display
- Wi-Fi Module (ESP32 built-in)
- ThingSpeak Cloud Platform
- Jumper wires
- Wokwi simulator

Circuit Diagram:



Component	ESP32 Pin
TRIG	GPIO 5
ECHO	GPIO 18
LED	GPIO 25
Buzzer	GPIO 26
LCD SDA	GPIO 21
LCD SCL	GPIO 22

Working Principle:

- Ultrasonic sensor measures object distance
- ESP32 reads the distance values
- If distance is below threshold, intruder is detected
- LED and buzzer activate as alert
- LCD displays intruder status
- Sensor values are uploaded to ThingSpeak cloud for monitoring

Code:

```
#include <WiFi.h>
#include "ThingSpeak.h"
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

#define TRIG_PIN 5
#define ECHO_PIN 18
#define LED_PIN 25
#define BUZZER_PIN 26

const char* ssid = "Wokwi-GUEST";
const char* password = "";

unsigned long channelID = 3420129;
const char* writeAPIKey = "3M59NTL79RQ5Y89O";

WiFiClient client;
LiquidCrystal_I2C lcd(0x27, 16, 2);

float readDistance() {
    digitalWrite(TRIG_PIN, LOW);
    delayMicroseconds(2);

    digitalWrite(TRIG_PIN, HIGH);
    delayMicroseconds(10);
    digitalWrite(TRIG_PIN, LOW);

    long duration = pulseIn(ECHO_PIN, HIGH);
    float distance = duration * 0.034 / 2;
    return distance;
}

void setup() {
    Serial.begin(115200);

    pinMode(TRIG_PIN, OUTPUT);
    pinMode(ECHO_PIN, INPUT);
    pinMode(LED_PIN, OUTPUT);
    pinMode(BUZZER_PIN, OUTPUT);

    lcd.init();
    lcd.backlight();

    lcd.setCursor(0, 0);
    lcd.print("Security Node");
```

```

lcd.setCursor(0, 1);
lcd.print("Booting...");
delay(2000);
lcd.clear();

WiFi.begin(ssid, password);

lcd.print("Connecting...");
Serial.print("Connecting WiFi");

while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
}

Serial.println("\nWiFi Connected");

lcd.clear();
lcd.setCursor(0, 0);
lcd.print("WiFi Connected");
delay(2000);

ThingSpeak.begin(client);

lcd.clear();
}

void loop() {
    float distance = readDistance();
    int intruderStatus = 0;

    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("Dist:");
    lcd.print(distance, 1);
    lcd.print("cm");

    Serial.println("=====");
    Serial.print("Distance: ");
    Serial.println(distance);

    if (distance < 30) {
        intruderStatus = 1;

        digitalWrite(LED_PIN, HIGH);

        tone(BUZZER_PIN, 1000);
        delay(300);
    }
}

```

```

noTone(BUZZER_PIN);

lcd.setCursor(0, 1);
lcd.print("INTRUDER ALERT");

Serial.println("Intruder Detected");
} else {
intruderStatus = 0;

digitalWrite(LED_PIN, LOW);

lcd.setCursor(0, 1);
lcd.print("SAFE ZONE");

Serial.println("Safe Zone");
}

ThingSpeak.setField(1, distance);
ThingSpeak.setField(2, intruderStatus);

int response = ThingSpeak.writeFields(channelID, writeAPIKey);

if (response == 200) {
Serial.println("Cloud Upload Success");
} else {
Serial.print("Upload Error: ");
Serial.println(response);
}

Serial.println("=====");
Serial.println();

delay(20000);
}

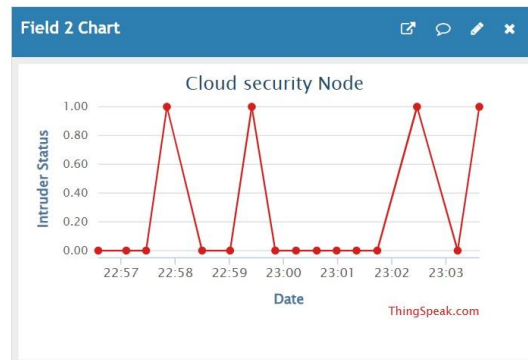
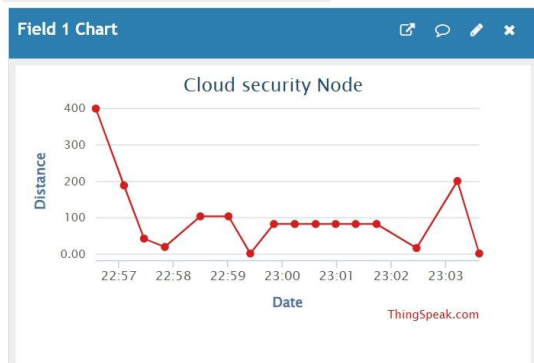
```

Output:

```
=====
Distance: 17.02
Intruder Detected
Cloud Upload Success
=====
```

```
=====
Distance: 200.92
Safe Zone
Cloud Upload Success
=====
```

```
=====
Distance: 1.97
Intruder Detected
Cloud Upload Success
=====
```



Result:

The cloud – connected security node was successfully implemented and tested in simulation. The system detected intruders and uploaded data to the cloud platform.

Conclusion:

Through this project, I learned ultrasonic sensor interfacing, cloud communication, Wi-Fi connectivity, and real-time security monitoring using ESP32.